

| Assessment | Test | Domain                    | Skill   | Difficulty                                          |
|------------|------|---------------------------|---------|-----------------------------------------------------|
| SAT        | Math | Geometry and Trigonometry | Circles | Medium <div><div></div><div></div><div></div></div> |

Question

A circle in the  $xy$ -plane has its center at  $(-4, 5)$  and the point  $(-8, 8)$  lies on the circle. Which equation represents this circle?

Answer

- A.  $(x - 4)^2 + (y + 5)^2 = 5$
- B.  $(x + 4)^2 + (y - 5)^2 = 5$
- C.  $(x - 4)^2 + (y + 5)^2 = 25$
- D.  $(x + 4)^2 + (y - 5)^2 = 25$

Correct Answer: D

Rationale

Choice D is correct. A circle in the  $xy$ -plane can be represented by an equation of the form  $(x - h)^2 + (y - k)^2 = r^2$ , where  $(h, k)$  is the center of the circle and  $r$  is the length of a radius of the circle. It's given that the circle has its center at  $(-4, 5)$ . Therefore,  $h = -4$  and  $k = 5$ . Substituting  $-4$  for  $h$  and  $5$  for  $k$  in the equation  $(x - h)^2 + (y - k)^2 = r^2$  yields  $(x - (-4))^2 + (y - 5)^2 = r^2$ , or  $(x + 4)^2 + (y - 5)^2 = r^2$ . It's also given that the point  $(-8, 8)$  lies on the circle. Substituting  $-8$  for  $x$  and  $8$  for  $y$  in the equation  $(x + 4)^2 + (y - 5)^2 = r^2$  yields  $(-8 + 4)^2 + (8 - 5)^2 = r^2$ , or  $(-4)^2 + (3)^2 = r^2$ , which is equivalent to  $16 + 9 = r^2$ , or  $25 = r^2$ . Substituting  $25$  for  $r^2$  in the equation  $(x + 4)^2 + (y - 5)^2 = r^2$  yields  $(x + 4)^2 + (y - 5)^2 = 25$ . Thus, the equation  $(x + 4)^2 + (y - 5)^2 = 25$  represents the circle.

Choice A is incorrect. The circle represented by this equation has its center at  $(4, -5)$ , not  $(-4, 5)$ , and the point  $(-8, 8)$  doesn't lie on the circle.

Choice B is incorrect. The point  $(-8, 8)$  doesn't lie on the circle represented by this equation.

Choice C is incorrect. The circle represented by this equation has its center at  $(4, -5)$ , not  $(-4, 5)$ , and the point  $(-8, 8)$  doesn't lie on the circle.